ADVANCED WEB TECHNOLOGIES · PROJECT

EcoTrace

Carbon-smart shipping decisions for small businesses

Measure, compare, forecast and reduce the CO_2 of every shipment

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React · Node.js · Express · MongoDB · Python ML

What we will cover

01



The problem

Why small businesses need carbon data for their shipments

02



Our solution

Objectives, workflow, architecture and the nine modules

03



Machine learning

Six models and how they performed against baselines

04



Technology

The full stack, from React to the Python ML service

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Feasibility

Technical, data and economic feasibility, and our conclusion

01

The problem

Small businesses are being pulled into carbon reporting without the tools to do it

Background: carbon reporting is reaching small suppliers



Freight is a major emitter

Freight transport is one of the largest sources of greenhouse-gas emissions.



MSMEs move goods mostly by road

Millions of Indian micro, small and medium enterprises ship raw materials and finished goods every day, mostly by road, the most carbon-intensive option after air.



Big clients now ask for emission data

Listed companies report under SEBI's Business Responsibility and Sustainability Reporting (BRSR) framework and are asking their suppliers for value-chain emissions.

The gap

Small businesses must supply carbon data but have no practical tool to measure or reduce their transport emissions.

Four core problems

No visibility

A small manufacturer does not know how much CO_2 each shipment, route or vendor generates. Records sit in spreadsheets and are never analysed.

Decisions on cost alone

Vendors and modes (road, rail, sea, air) are picked by price or habit. Nothing shows carbon, cost and time side by side.

No foresight

Businesses cannot predict next quarter's emissions, or which shipments will be late in the monsoon and festival seasons.

Existing tools do not fit

Enterprise platforms are expensive and complex. Free calculators handle one shipment at a time, with no history, comparisons or predictions.

“

FORMAL PROBLEM STATEMENT

Small and medium businesses lack an affordable, easy-to-use system to measure, compare, forecast and reduce the carbon emissions of their shipments.

As a result, logistics decisions ignore environmental impact, emission targets cannot be planned, and suppliers cannot provide the emission data their clients increasingly demand.

Who is affected



MSME owners

Cannot answer a client who asks for the carbon footprint of their deliveries.



Logistics managers

Choose vendors and modes without data on emissions or delay risk.



Large client companies

Need supplier (value-chain) emission data for ESG reporting.



Environment and society

Bear the avoidable emissions of inefficient, carbon-heavy transport choices.

02

Our solution: EcoTrace

A full-stack platform that turns shipment records into greener, cheaper and more reliable decisions

Six objectives

Calculate

The CO_2 of every shipment automatically, from route distance, weight and transport mode.

Visualise

Emissions through an interactive dashboard, route map and supply-chain view.

Compare

Vendor and mode options on carbon, cost and time, with adjustable priorities.

Predict

Future emissions and delay risk with machine learning, and detect abnormal shipments.

Optimize

Recommend the lowest-cost shipping plan that stays within a monthly CO_2 budget.

Report

Generate shareable monthly sustainability reports and send budget alerts.

Every shipment gets its CO_2 calculated automatically

EcoTrace is a web platform where a business logs or imports its shipments. It calculates their carbon footprint, shows where emissions come from, and uses machine learning and optimization to recommend better shipping decisions.

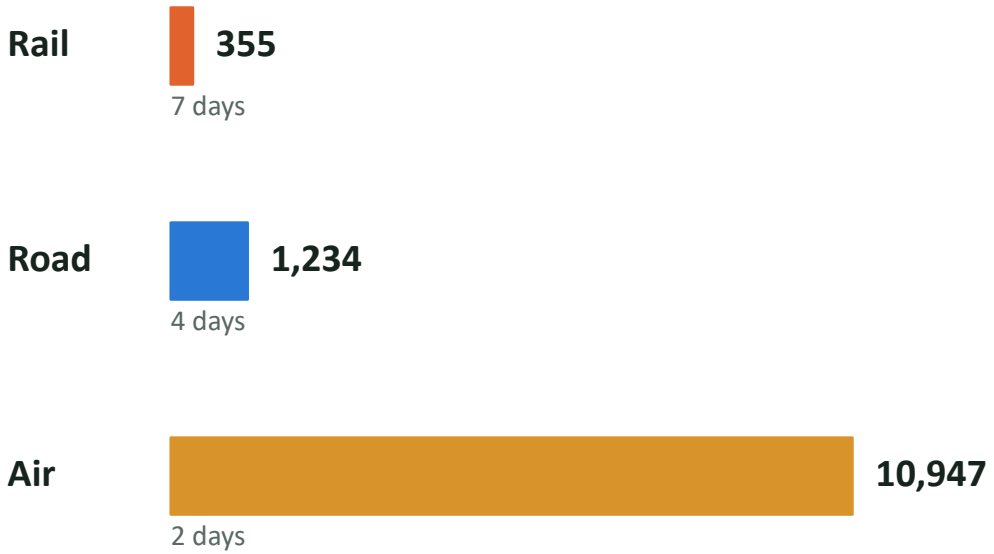
CO_2 = distance × weight × factor

Distance from OpenRouteService (truck profile). Emission factor in kg CO_2e per tonne-km, from the UK Government GHG Conversion Factors.

Mode	kg CO_2e / tonne-km
Road	0.107
Rail	0.028
Sea	0.016
Air	1.13

Mumbai → Delhi, 8 tonnes

kg CO_2e by transport mode



Air emits ~31× rail for the same load; rail takes 3 days longer than road.

How it works: eight steps

1

Register

Owner creates the company account (JWT, roles)

2

Set up

Add vendors, factory, warehouses and customers

3

Log shipments

One at a time or bulk CSV import

4

Auto-calculate

Route distance, CO_2, cost and transit days

5

Analyse

Dashboard by mode, vendor, route and month

6

Compare

Rank every vendor + mode option

7

Predict

Forecast 3 months; flag delay risk

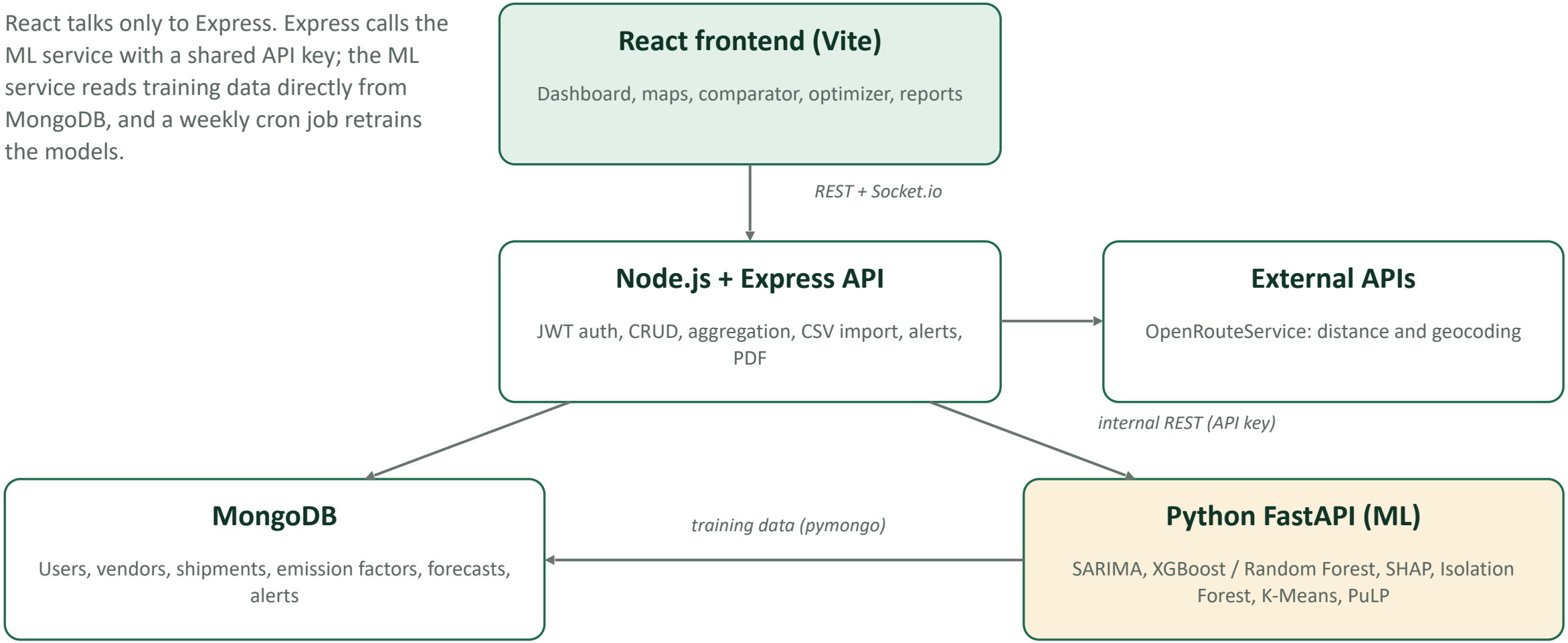
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Optimize & report

Cheapest plan under budget; PDF and alerts

System architecture

React talks only to Express. Express calls the ML service with a shared API key; the ML service reads training data directly from MongoDB, and a weekly cron job retrains the models.



Nine system modules, all working

Authentication & roles

JWT, bcrypt, Express middleware

Vendors & locations

React forms, REST API, MongoDB

Shipment logger

CSV import (Multer, csv-parser), ORS

Analytics dashboard

Recharts, React-Leaflet, aggregation

Option comparator

React state, Express scoring API

What-if simulator

Live re-calculation in React

Supply-chain graph

React Flow

Alerts & reports

Nodemailer, node-cron, pdfkit

Real-time updates

Socket.io

03

Machine learning

Six models that forecast, predict, explain, detect, segment and optimize

Six ML components in the Python service

Feature	Purpose	Model / library	User sees
Emission forecasting	Monthly emissions for the next 3 months	SARIMA with Indian season effects (Prophet if installed)	Forecast line with 80% band
Delay-risk prediction	Probability a shipment arrives late	Random Forest vs XGBoost	Green / amber / red badge
Explainable risk	Why a shipment is risky	SHAP	“Vendor late 8/10 times +21%”
Anomaly detection	Unusual or wrongly entered shipments	Isolation Forest	Flagged rows with a reason
Vendor segmentation	Group vendors by carbon, cost, reliability	K-Means	“Green and Reliable”
Carbon-budget optimizer	Cheapest mix within a CO_2 budget	PuLP linear programming	Recommended allocation

Results against our success criteria

37%

lower forecast error

MAE 11.8 t vs 18.6 t for the naive “same as last month” baseline

0.761

delay-model ROC-AUC

vs 0.680 for the rule baseline; F1 0.564 vs 0.535 on 965 held-out shipments

17.8%

cheaper shipping plan

and 60.8% less CO_2 than the actual Sept 2026 mix, within budget

14 / 14

bad records caught

every deliberately corrupted shipment, with no false alarms

80 ms

dashboard query

for 2 years (~4,900 shipments); target was under 2 seconds

Exact

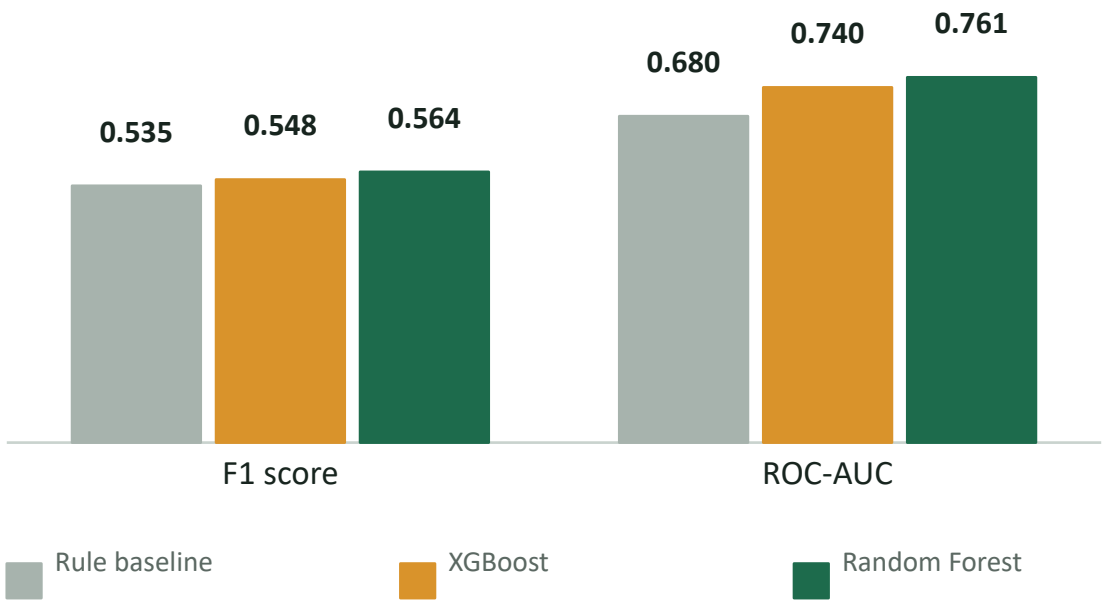
CO_2 calculation

136.8 km × 8 t × 0.107 = 117.10 kg, matched by the system

Models beat their baselines on held-out data

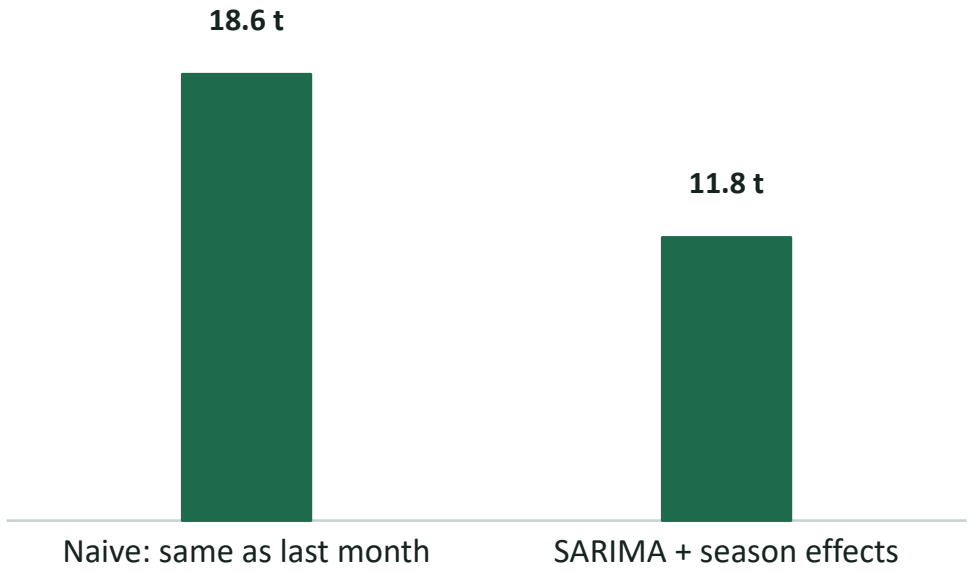
Delay-risk prediction

965 held-out shipments · higher is better



Emission forecast error (MAE)

Jul–Sep 2026 backtest · lower is better



What the website includes

Emissions dashboard

KPIs, monthly CO_2 by mode, carbon budget, forecast, route map and high-risk shipments

Shipments, vendors and locations

Log shipments or import CSV, with a live preview of distance, CO_2, cost, transit days and delay risk

Option comparator and what-if

Rank every vendor + mode with carbon / cost / time sliders; simulate shifting freight between modes

Carbon-budget optimizer

The cheapest shipping plan that meets demand and deadlines within a CO_2 budget

Supply-chain graph

Vendors → factory → customers, with links sized and coloured by CO_2

ML insights, reports and alerts

Forecasts, SHAP risk reasons, anomalies, vendor segments, monthly PDF reports and budget emails

Plus: JWT login with Owner, Logistics Manager and Admin roles, light and dark themes, and live updates between teammates (Socket.io).

04

Technology

The full stack, from HTML and React to MongoDB and a Python ML service

The full stack

Frontend

- HTML5, CSS3, JavaScript
- React + Vite
- React Router, Context API
- Axios
- Recharts
- React-Leaflet + OSM
- React Flow
- Socket.io client

Backend

- Node.js, Express
- JWT, bcrypt
- Multer, csv-parser
- Nodemailer
- node-cron
- pdfkit
- Socket.io
- dotenv

Database

- MongoDB (Atlas)
- Mongoose schemas
- Aggregation pipelines
- Geospatial indexes
- Distance cache

Machine learning

- FastAPI + Uvicorn
- pandas, NumPy
- scikit-learn
- XGBoost
- statsmodels SARIMA
- SHAP
- PuLP
- joblib, pymongo

APIs & hosting

- OpenRouteService
- OpenStreetMap tiles
- UK GHG factors
- Vercel / Netlify
- Render / Railway
- Postman, VS Code

05

Feasibility

Technically, economically and practically achievable in one semester

Technical, data and economic feasibility

Component	Difficulty
React UI, charts, maps	Medium
Express REST API + JWT	Medium
MongoDB schemas, aggregations	Medium
CO_2 calculation	Easy
Routing API integration	Easy
Forecasting & delay models	Medium
Optimizer (PuLP)	**Medium-High**
Node ↔ Python integration	Medium

Mature, free, well-documented libraries; no new research needed.

₹0

running cost

MongoDB Atlas M0 · OpenRouteService free key · Vercel / Netlify · Render / Railway free tier · open-source libraries



Data feasibility

- Emission factors are public (UK GHG Conversion Factors), stored locally
- Synthetic generator: 2 years with monsoon delays, festival peaks, late vendors
- Real businesses can import their own CSV records

CONCLUSION

EcoTrace gives small businesses an affordable way to understand and reduce the carbon footprint of their shipments.

It brings together the full scope of Advanced Web Technologies (React, Node/Express REST, MongoDB, third-party APIs, real-time communication) with meaningful machine learning and optimization in Python.

Thank you

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Try it: owner@ecotrace.demo / demo1234